



Time-Series-Driven Predictive Maintenance with NASA C-MAPSS

**Short-horizon failure-risk modeling and maintenance policy
comparison on FD001**

MA Statistics

Chun-Kai Hsu



Agenda

- 1 Problem and motivation
- 2 Dataset and setup
- 3 Time-series features and models
- 4 Diagnostics and predictive results
- 5 Maintenance policy comparison
- 6 Conclusions and limitations



Introduction and Motivation

Motivation

- Unexpected failures are **expensive**.
- Replacing too early is also expensive.
- **Predictive maintenance tries to intervene before failure**, but not too early.

Why This Is a Time-Series Problem

Problem

- Degradation develops across cycles.
- Sensor levels, slopes, and persistence all matter.
- Static snapshots miss important temporal information.

1. Raw sensor stream
2. Temporal summaries
3. Failure risk estimate

Research Question

Prediction target

- failure within the next 30 cycles

Decision target

- lower maintenance cost using predicted risk

Main comparison

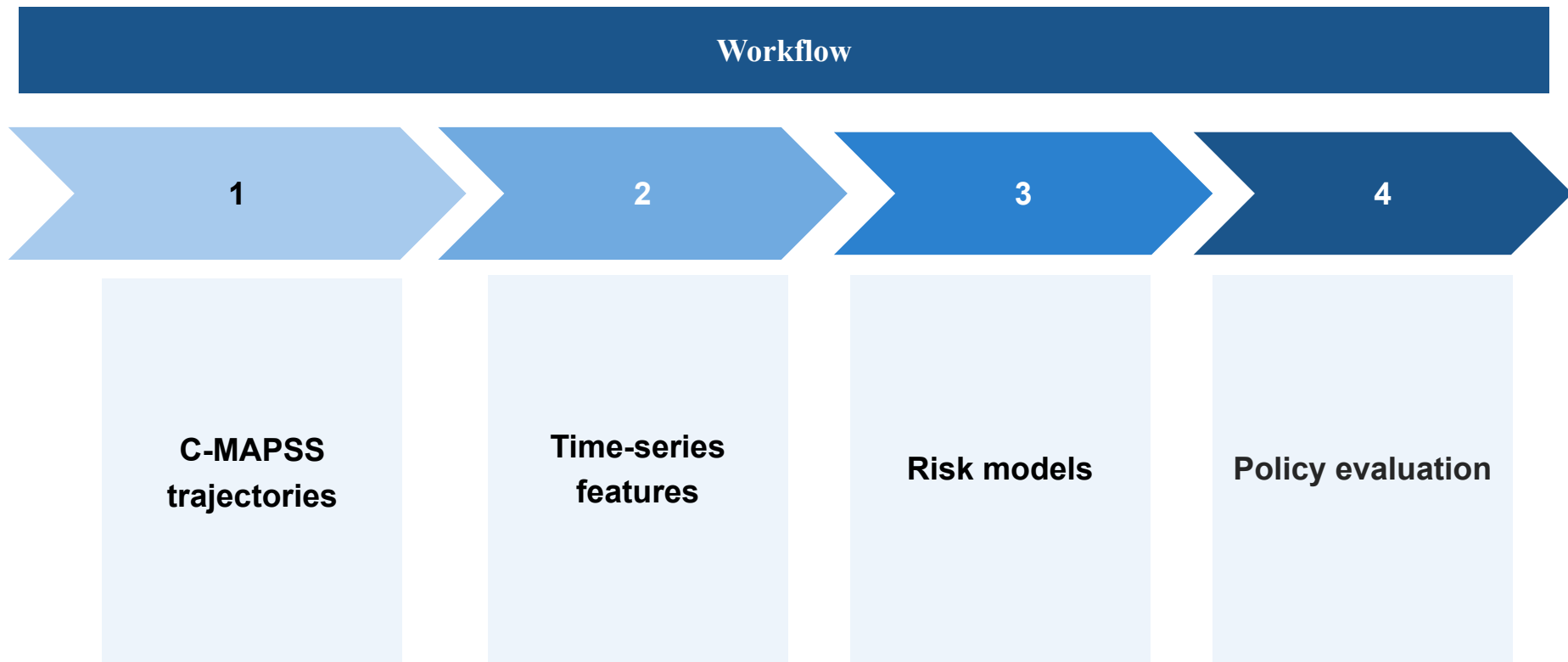
- snapshot-only model versus time-series-aware models

NASA C-MAPSS FD001 Dataset

- Run-to-failure turbofan engine trajectories
- 3 operating settings and multiple sensor channels
- I use engine-unit splits from `train_FD001.txt`
- This keeps complete trajectories for policy evaluation

Split	Machines	Mean Failure Time	Positive Rate within 30 cycles
Train	70	207.4	0.149
Validation	15	200.1	0.155
Test	15	207.3	0.150

Workflow Overview



Time-Series Features and Predictive Models

Feature families

- rolling means and standard deviations
- rolling slopes
- lagged values
- first and second differences
- exponentially weighted moving averages
- Holt level and trend summaries
- short-run versus long-run sensor contrasts

Models

baseline_glm

- current-cycle settings and sensors only

advanced_glm

- snapshot + temporal summaries

nonlinear_ts_forest

- **richer nonlinear temporal model**

equation

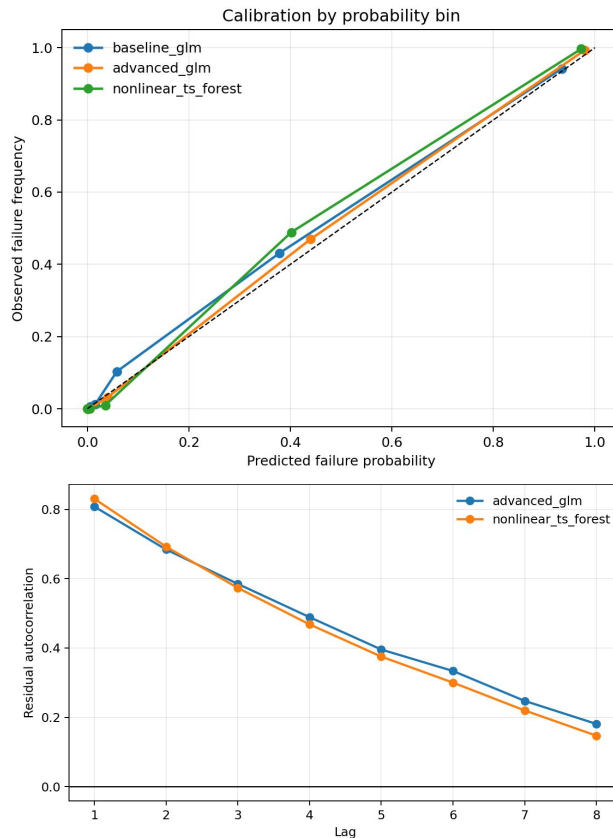
- **$P(\text{Fail within 30 cycles at time } t \mid \text{features at time } t)$**

code

- baseline_glm = current settings + current sensors
- advanced_glm = baseline + rolling summaries + Holt features
- **nonlinear_ts_forest = richer lags + differences + EWMS + nonlinear interactions**

Diagnostics and Validation

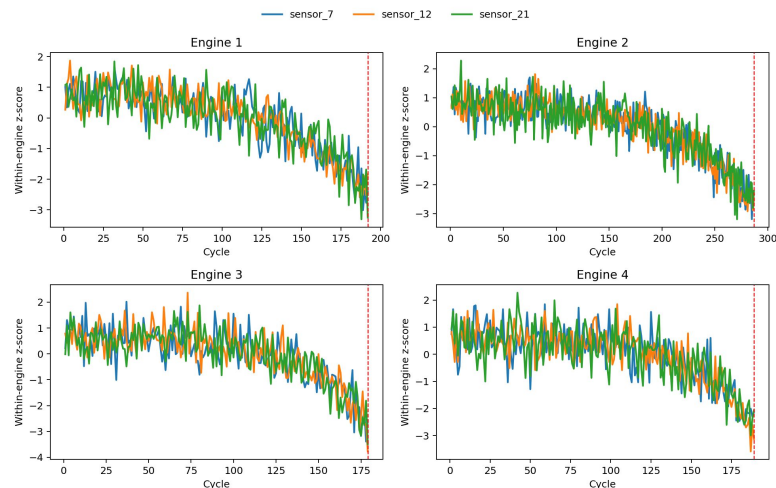
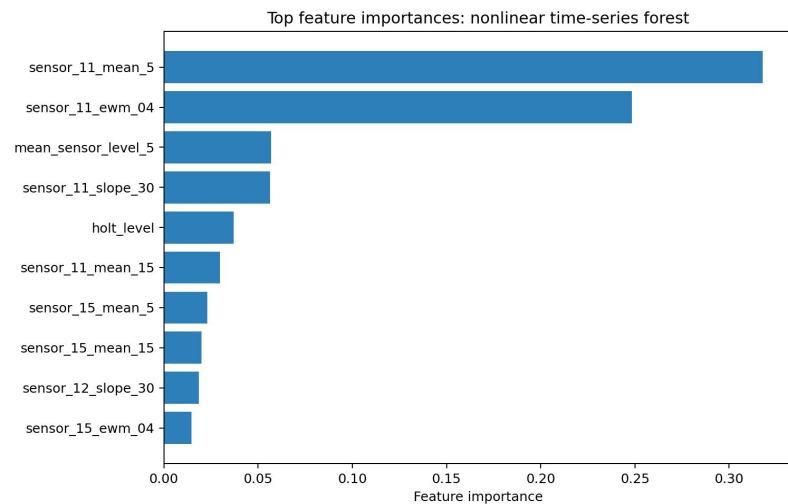
- stationarity after differencing
- probability calibration
- residual autocorrelation



Series	ADF Statistic	p-value
sensor_7 diff	-40.89	0.000
sensor_12 diff	-48.51	0.000
sensor_21 diff	-20.74	0.000

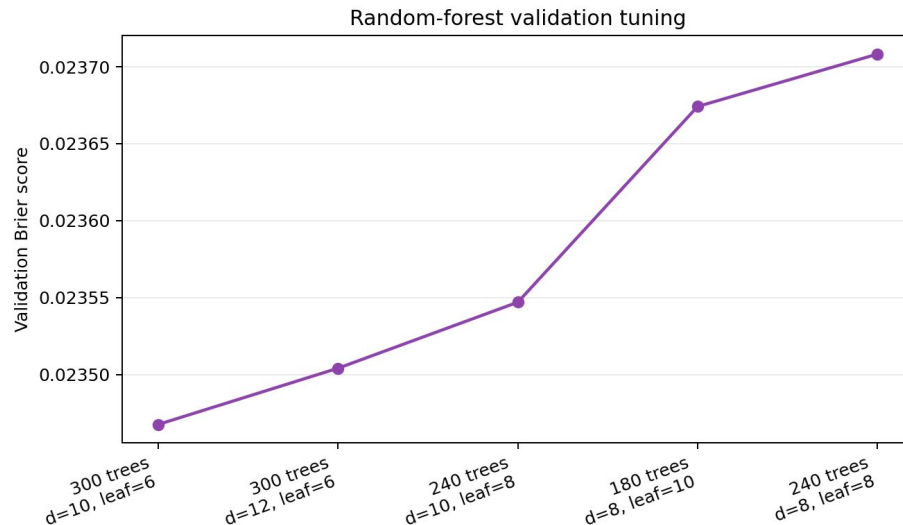
Predictive Results

Model	AUC	Brier	Log Loss
nonlinear_ts_forest	0.9949	0.0221	0.0738
advanced_glm	0.9936	0.0227	0.0739
baseline_glm	0.9808	0.0371	0.1234



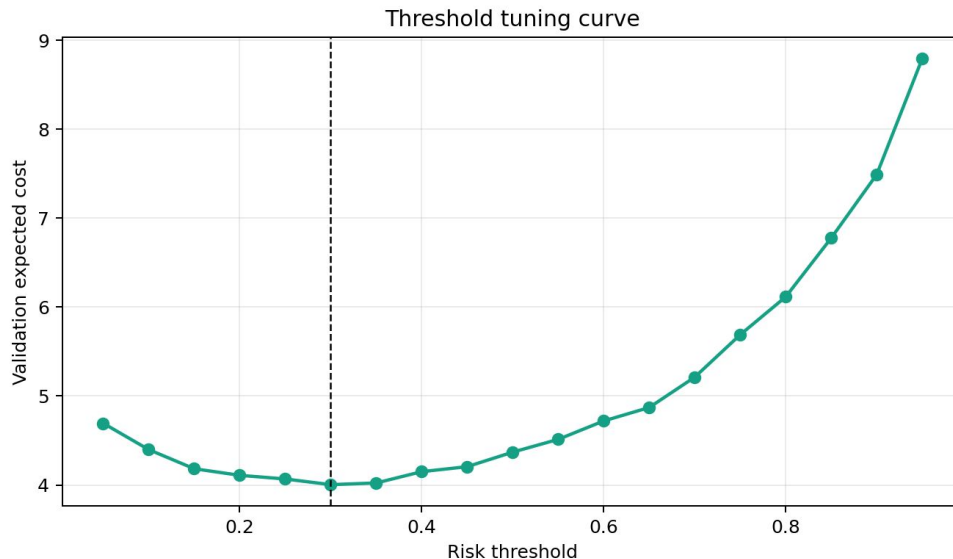
Nonlinear Model Tuning

- small validation-based hyperparameter sweep
- tuning criterion: lowest validation Brier score
- selected forest: 300 trees, max depth 10, min leaf 6, max features 0.5



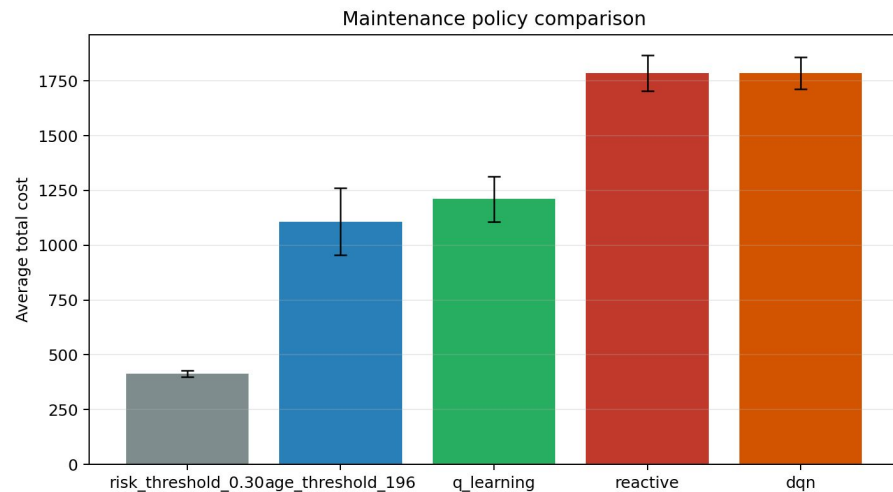
Risk Threshold Tuning

- validation-based threshold sweep
- objective: **minimize expected maintenance cost**
- selected threshold: **0.30**



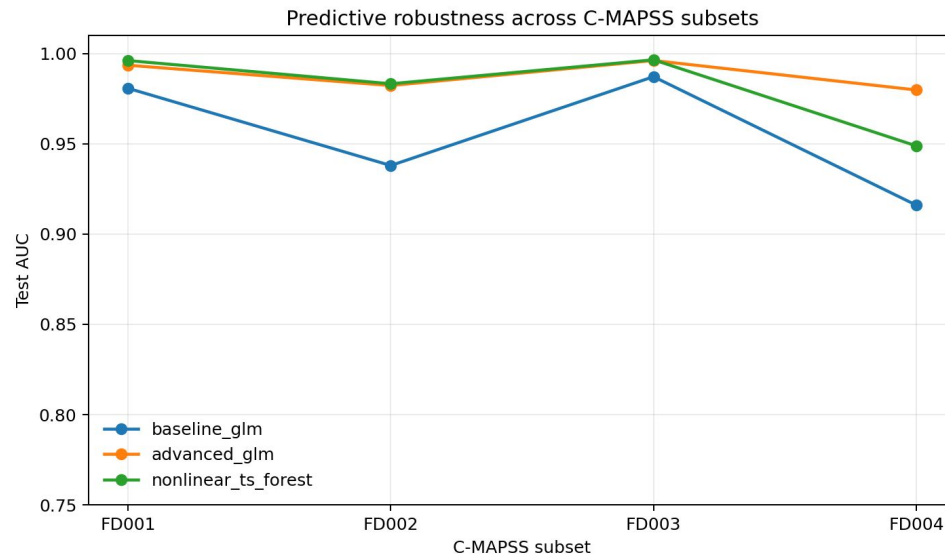
Maintenance Policy Comparison

Policy	Average Cost	Average Replacements	Average Failures
risk_threshold_0.30	417.9	23.2	0.0
age_threshold_196	1108.2	20.9	9.5
reactive	1784.6	18.8	18.8
q_learning	1210.9	22.36	10.5
dqn	1784.6	18.8	18.8



Robustness Across FD002-FD004

- appendix-style predictive extension
- same snapshot-vs-temporal comparison on all four C-MAPSS subsets
- advanced temporal model remains strong across harder subsets



Conclusions

Interpretation

- **Temporal context improves over a snapshot-only baseline.**
- Rolling and smoothed features carry strong predictive signal.
- Better risk estimates lead to much better maintenance decisions.
- Simple threshold control beats controller complexity here.

Limitations and Future Work

- Only the `FD001` subset is used
- The policy layer is intentionally lightweight
- **Full policy experiments are only run on FD001**
- Future work: regime-aware models, recurrent methods, state-space models

Conclusion

- **Time-series features outperform a snapshot-only baseline** on FD001.
- The strongest predictive model is `nonlinear_ts_forest`.
- Accurate risk estimates lead to **much lower maintenance cost**.
- **A simple threshold policy** is the strongest practical controller here.



Time-Series-Driven Predictive Maintenance with NASA C-MAPSS

**Short-horizon failure-risk modeling and maintenance policy
comparison on FD001**

Q&A

